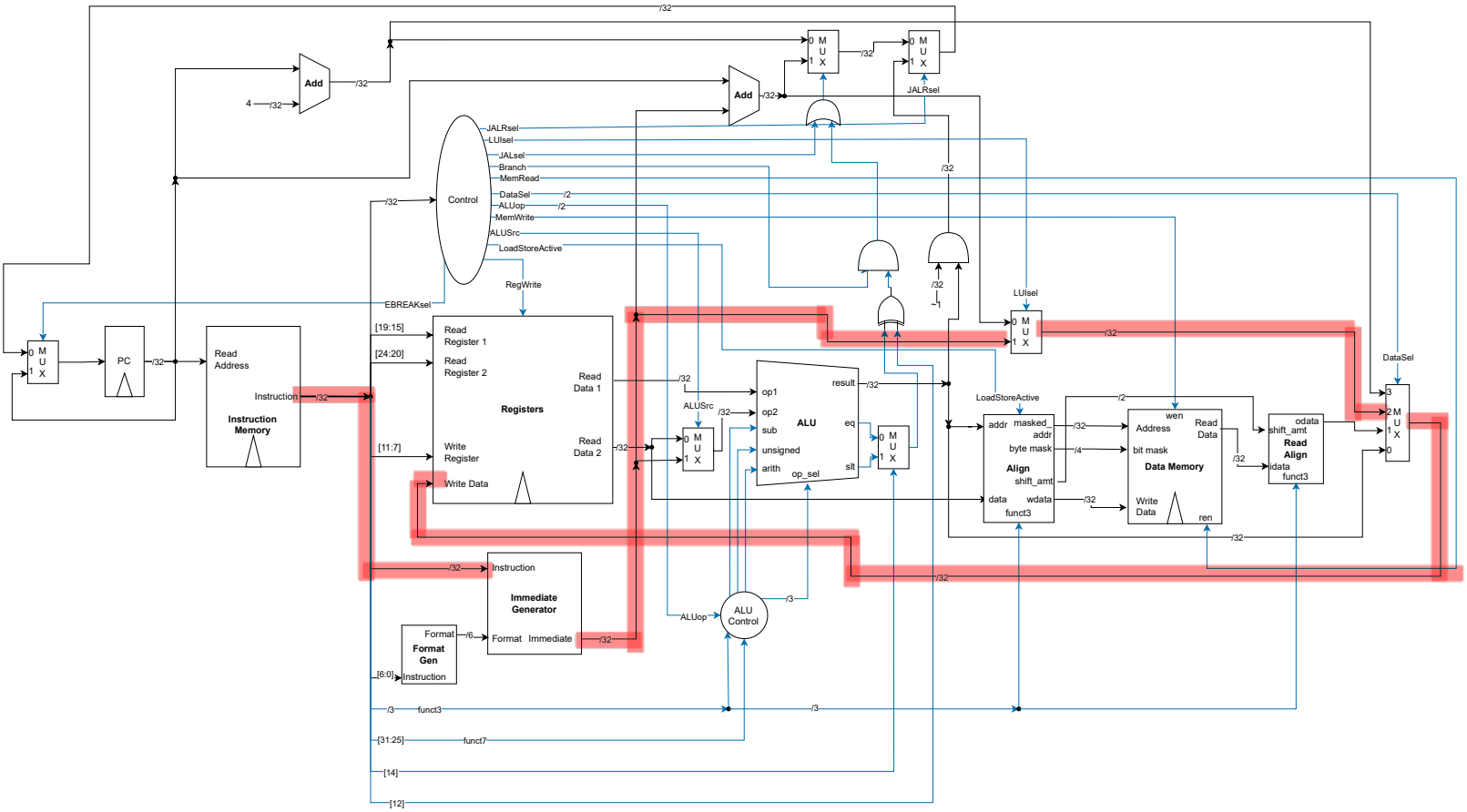

Add
4
/32
JALRsel
LUIsel
JALsel
Branch
MemRead
DataSel
/2
ALUop
MemWrite
ALUSrc
LoadStoreActive
RegWrite
EBREAKsel
Control
PC
Read Address
Instruction
Instruction Memory
[19:15]
[24:20]
[11:7]
Read Register 1
Read Register 2
Write Register
Write Data
Read Data 1
Read Data 2
Registers
ALUSrc
op1
op2
sub
unsigned
arith
op_sel
ALU
result
eq
slt
ALUop
ALU Control
Instruction
Immediate Generator
Format
Immediate
Format Gen
[6:0]
/6
/3
funct3
[31:25]
funct7
[14]
[12]
addr
masked addr
byte mask
Align
shift_amt
data
wdata
funct3
wen
Address
Read Data
bit mask
Data Memory
Write Data
ren
/4
odata
shift_amt
Read Align
idata
funct3
DataSel
0
1
2
3